

XCM/XFA V2 – Structured Listing Integration Proposal

1. Executive Summary

The XCM/XFA V2 platform is being rebuilt to transition from a manual PDF/email listing workflow to a structured, auditable integration model. The objective is not direct ALTEA integration, but a clean and secure boundary where KLM may ingest structured listing data electronically.

2. Current Operational Model

Current flow: XCM/XFA generates PDF → Email to station → Manual input into ALTEA DCS.

3. Proposed Tier 2 Integration Model

Proposed flow: XCM/XFA generates structured JSON payload + PDF artifact → Secure transmission to KLM ingestion endpoint → KLM internal service validates and inserts into ALTEA → Response returned with security numbers/booking references.

4. Architectural Principles

- No direct ALTEA access required by XCM/XFA.
- Delivery channel abstraction (Email, HTTP JSON, SFTP).
- Versioned JSON schema for forward compatibility.
- Immutable batch artifacts (PDF + JSON) stored for audit.
- Delivery status logging and retry control.
- Future-ready acknowledgement handling for returned security numbers.

5. Delivery Channels (Future-Proof Design)

- EMAIL_PDF – Current operational model.
- HTTP_JSON – Secure POST to KLM endpoint (recommended future state).

- SFTP_JSON / CSV – Alternative structured transfer.
- NONE – Artifact generation without delivery (testing mode).

6. Example JSON Payload Structure (Conceptual)

Schema versioned, batch-based payload including airport, departure date, flight details, passenger identity, priority, inbound reference, and passport information where policy permits.

7. Governance & Security

- Role-based admin access.
- Per-airport delivery policy stored in database.
- 90-day artifact retention policy.
- Full audit trail of batch generation and delivery.
- Optional payload signing or token-based authentication.

8. Strategic Positioning

The XCM/XFA V2 platform positions itself as an external structured integration layer that reduces manual workload while allowing KLM to maintain full operational control over ALTEA and internal systems.